Strike Three, You're Out?

A Mathematical Analysis of How an Automated Strike Zone Could Alter Major League Baseball Hitter Performance

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Introduction

- In 2018, Boston University found Major League Baseball (MLB) umpires missed an **average of 14 pitch calls per game.**
- Along with this, **55 games** were found to end on an incorrect pitch call

What Can Be Done?

Seeing the opportunity for improvement, MLB applied advanced technology to solve this issue

Specifically, an Automated Ball and Strike (ABS) Challenge System will be implemented full time in the MLB starting in 2026.

What is ABS?

Two versions of ABS exist

- **Full ABS**
  - The umpire's job is automated; **no human judgment involved in this system**
- **ABS Challenge System**
  - Umpire's job is essentially same as before; **however there is an opportunity for players to question a possible mistake**

Initial Testing

- Full ABS was first introduced in the Atlantic League in 2019 (independent affiliate of MLB)
- The Florida State League (lowest level of minor league baseball) was the first MLB affiliate to experience both ABS systems in action
- After this initial experimentation, the ABS Challenge System was found to be more suitable for players, fan experience, and pace of play

Benefits of ABS Challenge System

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- Strike zone definition is rigid; not nearly as much inconsistency with the dimensions of the strike zone

ABS Strike Zone Dimensions

Figure: Strike Zone Dimensions designed by ABS system

Triple-A Testing

Initial testing did not end in the low-level MLB affiliates

In 2023, **both ABS systems were used in Triple-A** (highest level of Minor League Baseball) to ensure player ability wasn't a major factor in preference of either system

Triple-A Testing

Throughout the 2023 season, Triple-A games from Tuesday-Thursday used Full ABS and weekend games used the Challenge System

As before, ABS Challenge System was preferred due to the removal of the Full ABS implementation in the middle of the 2024 Triple-A season.

Note: The following study treats both systems as equal for larger sample sizes and robust analyses.

Study Focus

Research Question: Will the implementation of the ABS system in the MLB affect standard batting statistics, such as, strikeout percentage, walk rate, swinging strike percentage, and the number of balls, called strikes, and pitches seen throughout a season?

Data Collection

- Data was collected from FanGraphs on all of the hitters who recorded at least 200 plate appearances in a given season over the years 2019-2025.
- Data set contains 2,068 observations, containing **1,135 distinct players**
- Other variables included
  - Age
  - Plate Appearances
  - Team
  - Season

Data Sample

	Season	Name	Team	Level	Age	PA	BB.
10642-2022	2022	Carlos Perez	COL	AAA	31	522	0.10344828
10642-2024	2024	Carlos Perez	OAK	AAA	33	468	0.10470085
10642-2025	2025	Carlos Perez	CHC	AAA	34	449	0.10913140
10655-2019	2019	Rob Brantly	PHI	AAA	29	272	0.11764706
10655-2021	2021	Rob Brantly	NYY	AAA	31	264	0.07575757
10655-2022	2022	Rob Brantly	NYY	AAA	32	201	0.05472637
10700-2019	2019	Ryan LaMarre	ATL	AAA	30	455	0.08351648
10700-2021	2021	Ryan LaMarre	NYY	AAA	32	240	0.11250000
10711-2019	2019	Arismendy Alcantara	NYM	AAA	27	339	0.09439528
10711-2021	2021	Arismendy Alcantara	SFG	AAA	29	255	0.08235294

Figure: Sample of the data collected

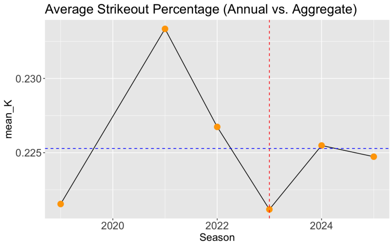

Figure: Comparisons of annual vs aggregate statistics for strikeout percentage

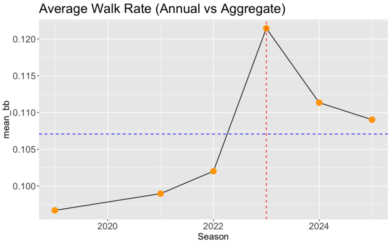

Figure: Comparisons of annual vs aggregate statistics for walk rate

Initial Observations

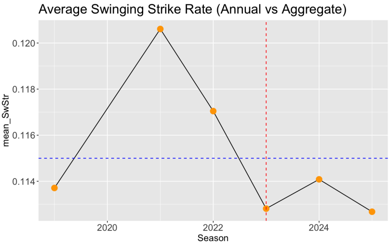

Figure: Comparisons of annual vs aggregate statistics for swinging strike percentage

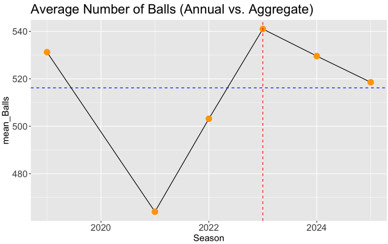

Figure: Comparisons of annual vs aggregate statistics for number of balls

Initial Observations

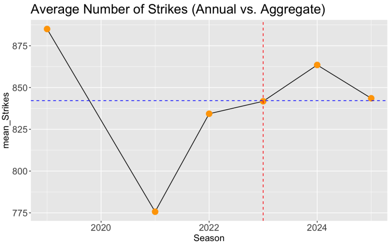

Figure: Comparisons of annual vs aggregate statistics for number of called strikes

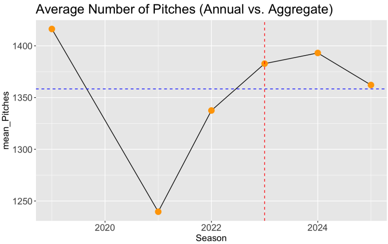

Figure: Comparisons of annual vs aggregate statistics for number of pitches

Hypothesis Tests

- To investigate this common trend, I began by using hypothesis tests to compare the mean for each metric in two time periods: **Before ABS (2019-2022) and During ABS (2023-2025)**

- A Welch t-test (rate variables) and two sample z-tests (count variables) were used to investigate the evident change in the six metrics

Hypothesis Tests

Table 1: Comparison of Batting Metrics Before and During ABS

Batting Metric	**Mean (Before)**	**Mean (During)**	**Difference**	**Test Statistic**	**95% CI**
Strikeout percentage	0.2269	0.2238	0.0031	1.2085	$(-0.0019, 0.0084)$
Walk Rate	0.0993	0.1141	-0.0148	-10.186	$(-0.0177, -0.0120)$
Swinging Strike percentage	0.1170	0.1132	0.0038	2.6181	$(0.00095, 0.0066)$
Balls	500.86	530.06	-29.2	-4.0877	$(-43.21, -15.19)$
Strikes	833.82	849.68	-15.85	-1.4487	$(-37.32, 5.61)$
Pitches	1334.68	1379.74	-45.05	-2.5476	$(-79.73, -10.37)$

Hypothesis Test Results

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- The swinging strike percentage **was slightly higher before the introduction of ABS**, indicating that hitters may have been better able to make contact under the new system.

- The number of balls and total pitches per season **both rose after the implementation of ABS**. This trend suggests that hitters may be experiencing longer at-bats and showing greater patience at the plate, possibly due to a tighter and more consistent strike zone.

Further Investigation

The hypothesis tests and knowing a gradual readjustment is present for each metric led me to perform **yearly effect analysis**.

Due to the structure of my data, I decided to move forward with a **Fixed Effects Regression** model.

Fixed Effects Regression

A **Fixed Effects model** for panel data has the form

$$Y_{it} = \beta_1 X_{1,it} + \beta_2 X_{2,it} + \cdots + \beta_j X_{j,it} + \alpha_i + \epsilon_{it},$$

where Y_{it} is the value of the dependent variable for an individual i at time t. $X_{j,it}$ for $j = 1, 2, \ldots, n$ represents the independent variables in the regression model for an individual i at time t, with corresponding coefficients β_j for $j = 1, 2, \ldots, n$. α_i represents the control for individual fixed effects, variables that stay constant over time for each individual, and ϵ_{it} is the error term.

Fixed Effects OLS

The regression analysis on the dependent variables: strikeout percentage, walk rate, and swinging strike percentage was performed by **Fixed Effects Ordinary Least Squares (OLS)** estimation with the following regression equation

$$\begin{aligned} Y_{it} = {} & D_1 abs2023_{it} + D_2 abs2024_{it} + D_3 abs2025_{it} \\ & + B_1 Age_{it} + B_2 PA_{it} + \alpha_i + \epsilon_{it}. \end{aligned}$$

Fixed Effects OLS Models

Dependent Variables:	Strikeout Percentage	Walk Rate	Swinging Strike Percentage
Model:	(1)	(2)	(3)
Variables			
abs2023	-0.0106***	0.0162***	-0.0059***
	(0.0035)	(0.0024)	(0.0017)
abs2024	-0.0117**	0.0026	-0.0059**
	(0.0046)	(0.0031)	(0.0024)
abs2025	-0.0226***	5.6×10^{-5}	-0.0114***
	(0.0058)	(0.0039)	(0.0029)
Age	0.0044***	0.0017**	0.0012**
	(0.0013)	(0.0008)	(0.0006)
Plate Appearances	-9.04×10^{-6}	-1.21×10^{-5}*	-6.01×10^{-6}
	(1.07×10^{-5})	(7.21×10^{-6})	(4.95×10^{-6})
Fixed-effects			
PlayerId	Yes	Yes	Yes
Fit statistics			
Observations	1,500	1,500	1,500
R^2	0.80856	0.73876	0.84968
Within R^2	0.02040	0.11783	0.02666

Clustered (PlayerId) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

A **Fixed Effects Poisson model** measures the count of observations in the dependent variable, y_{it}, for an individual i at a time t. Fixed Effects Poisson models act as a discrete probability distribution with the following probability function

$$f(y_{it} \mid X_{it}, \alpha_i, \beta) = \frac{\exp(-\mu_{it})\, \mu_{it}^{y_{it}}}{y_{it}!}, \qquad i = 1, 2, \ldots, N; \; t = 1, 2, \ldots, T$$

where y_{it} is the dependent variable for individuals i at time t, X_{it} is the vector containing observations on the all independent variables, α_i represents the fixed effects that are controlled for over time, and β is a vector containing all of the regression coefficients.

Fixed Effects Poisson

From prior definitions of the Poisson distribution, we then can see that the regression equation is of the form

$$log(\mathbb{E}[y_{it}|X_{it}, \alpha_i, \beta]) = \alpha_i + X_{it}\beta$$

(Note: There is no error term due to the fact that the Poisson distribution already accounts for the error term in the randomness of the distribution)

Fixed Effects Poisson

Observe that the specific regression equation used for my analysis on the number of balls, strikes, and pitches seen in a season is

$$\begin{aligned}\log(\mathbb{E}[Y_{it}]) = D_1 abs2023_{it} + D_2 abs2024_{it} + D_3 abs2025_{it} \\ + B_1 Age_{it} + \alpha_i + \log(PA_{it})\end{aligned}$$

where D_i for $i = 1, 2, 3$ are the coefficients for the respective indicator variables, $\log(PA_{it})$ represents an offset, and α_i represents the fixed effects.

Fixed Effects Poisson

Fixed Effects Poisson Models

Dependent Variables:	Balls	Strikes	Pitches
Model:	(1)	(2)	(3)
Variables			
abs2023	0.0443***	-0.0161***	0.0070*
	(0.0075)	(0.0036)	(0.0038)
abs2024	-0.0004	-0.0067	-0.0044
	(0.0101)	(0.0046)	(0.0050)
abs2025	0.0011	-0.0116*	-0.0068
	(0.0129)	(0.0061)	(0.0066)
Age	0.0016	0.0004	0.0009
	(0.0028)	(0.0013)	(0.0014)
Fixed-effects			
PlayerId	Yes	Yes	Yes
Fit statistics			
Observations	1,500	1,500	1,500
Squared Correlation	0.97140	0.99249	0.99188
Pseudo R^2	0.83772	0.88880	0.92169
BIC	18,536.0	17,830.7	19,206.1

Clustered (PlayerId) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

At first glance, it may seem contradictory that the hypothesis tests and regression analyses tell **slightly different stories** about the impact of ABS.

It is also essential to consider the sample sizes underlying each indicator variable in the regressions; **cannot attribute direct causality to solely ABS**.

Future Considerations

With further data collection, a study like this could greatly benefit from employing a Difference-in-Differences (DiD) regression model.

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Future research could take advantage of the advanced technology in MLB stadiums to analyze more detailed data.

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